

About the project

The idea of the project is to analyze the social media usage among young people (13-25 year olds). The dataset has columns such as the phone usage, social media usage, anxiety score, depression score, sleep hours, etc. I thought this dataset is valuable to check the correlations and comparisons that I wanted to visualize. The idea comes from the personal experience I have and others may have when it comes to using social media and how much we use it can affect our day to day basis.

Preprocessing

For this step, I checked the different components of the dataset and analyze that specific columns would be of high use in the project and how I want to cover this topic. Therefore I dropped some columns from the original dataset and filter out any person older than 25.

```
: df.columns

: Index(['user_id', 'age', 'gender', 'daily_screen_time_hours',
        'phone_usage_hours', 'laptop_usage_hours', 'tablet_usage_hours',
        'tv_usage_hours', 'social_media_hours', 'work_related_hours',
        'entertainment_hours', 'gaming_hours', 'sleep_duration_hours',
        'sleep_quality', 'mood_rating', 'stress_level',
        'physical_activity_hours_per_week', 'location_type',
        'mental_health_score', 'uses_wellness_apps', 'eats_healthy',
        'caffeine_intake_mg_per_day', 'weekly_anxiety_score',
        'weekly_depression_score', 'mindfulness_minutes_per_day'],
        dtype='object')
```

```
df_yp = df[(df['age'] <= 25) & (df['gender'] != 'Other')]
df_yp = df_yp.copy()
df_yp[:5]
```

	user_id	age	gender	daily_screen_time_hours	phone_usage_hours	laptop_usage_hours	tablet_usage_hours	tv_usage_hours	social_media_hours	work_related_hours
5	user_6	20	Female	9.9	3.2	2.7	1.2	3.3	1.6	2.5
10	user_11	23	Female	6.6	2.4	1.0	1.3	0.6	2.3	1.8
11	user_12	23	Male	5.2	3.9	1.8	0.7	2.5	2.5	2.4
16	user_17	15	Male	4.2	2.4	3.1	0.0	2.1	1.1	0.9
18	user_19	14	Male	5.5	2.7	0.8	1.2	0.1	3.0	1.2

5 rows × 25 columns

After:

```
df_yp.columns
```

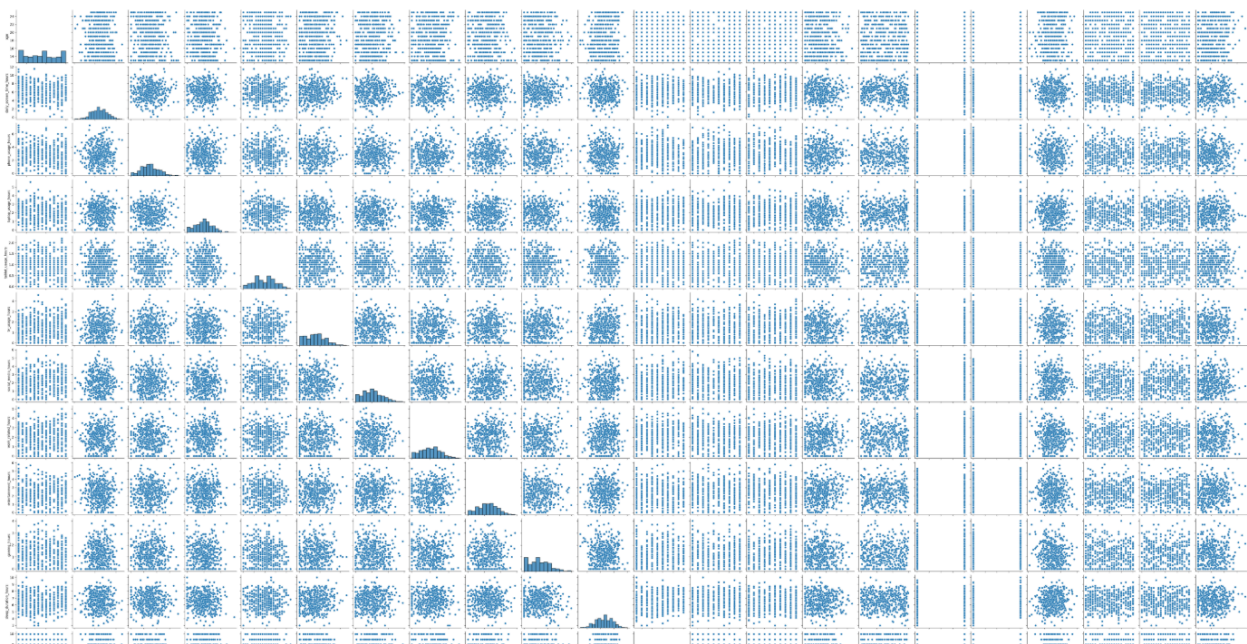
```
Index(['age', 'gender', 'daily_screen_time_hours', 'phone_usage_hours',  
      'social_media_hours', 'sleep_duration_hours', 'sleep_quality',  
      'mood_rating', 'stress_level', 'location_type', 'mental_health_score',  
      'caffeine_intake_mg_per_day', 'weekly_anxiety_score',  
      'weekly_depression_score'],  
      dtype='object')
```

Visualizations:

These are some of the visualizations that i wanted to use to first analyze the dataset and how it can be used when it comes to the approach I wanted to show:

```
sns.pairplot(df_yp, diag_kind='hist')
```

<seaborn.axisgrid.PairGrid at 0x12db6dc68d0>



This visualization was trying to analyze when it comes to depression and anxiety levels compare to phone usage and social media usage in young people and, based on the visualization, there seems to be a a peek on anxiety and depression with a social media usage of 2 hours

```
fig = make_subplots(rows=1, cols=2)

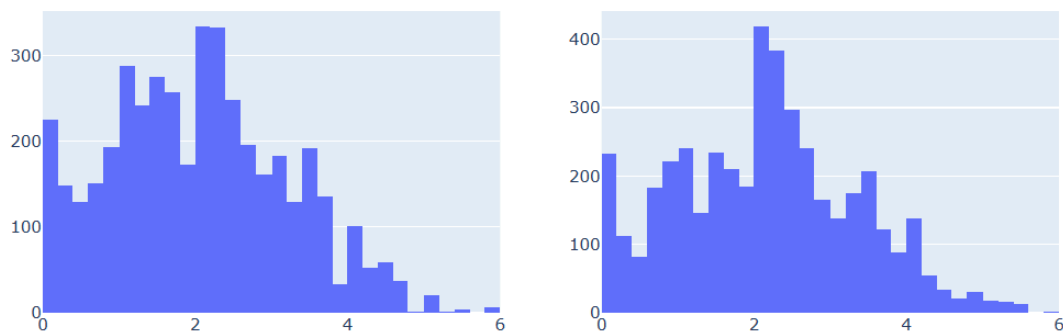
fig1 = px.histogram(df_yp, x="social_media_hours", y="weekly_anxiety_score",
                    title='Social Media Usage in relation to Weekly Anxiety')

fig2 = px.histogram(df_yp, x="social_media_hours", y="weekly_depression_score",
                    title='Social Media Usage in relation to Weekly Anxiety')

fig.add_trace(fig1.data[0], row=1, col=1)
fig.add_trace(fig2.data[0], row=1, col=2)

fig.update_layout(width=900, height=400, title_text='Depression and Anxiety levels in Correlation to Social Media Usage')
fig.show()
```

Depression and Anxiety levels in Correlation to Social Media Usage



Therefore I also analyze if there is a equal level of anxiety from female to male and therefore I wanted to check the anxiety levels compare to social media with the difference that I wanted to highlight how much of the data comes from female users vs male users. Which can show that females tend to feel more anxious over social media compared to males

```
fig=px.bar(df_yp, x='social_media_hours', y='weekly_anxiety_score', color='gender')
fig.show()
```

